

1 Reliable comparison of severe felt-intensity
2 classifiers with versioned and dependent event
3 data: a USGS DYFI benchmark

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9 **Abstract**

10 Engineers comparing classifiers on evolving event catalogues can reach false con-
11 clusions when related events cross data splits or small score gaps are treated as
12 wins. We test whether classifiers of severe felt intensity (event-maximum commu-
13 nity decimal intensity ≥ 6) follow fixed input and evaluation rules and whether
14 their gains remain supported under a fixed catalogue, event grouping, tempo-
15 ral holdout, and paired uncertainty. From a frozen USGS “Did You Feel It?”
16 (DYFI) snapshot of 7,520 $M \geq 5$ records, we formed 2,362 eligible events, held
17 out 430 events from 2023–2024, and compared fixed baselines with Brier score
18 and bootstrap intervals that resampled whole event sequences. All four learned
19 models—magnitude-only and magnitude-plus-depth logistic regression, random
20 forest, and XGBoost—improved on the constant-prevalence reference. Adding
21 depth to the magnitude-only logistic model reduced Brier score by 0.018 (95% CI
22 0.010–0.025). The magnitude-plus-depth logistic model and XGBoost both had
23 Brier 0.181 to three decimals, and their direct comparison remained unresolved.
24 The source-only input rule excluded a candidate using event year. A complemen-
25 tary random-forest diagnostic gave similar point performance when related events
26 were kept together or split randomly; without an interval, we report it descrip-
27 tively. The benchmark helps engineers reject inadmissible comparisons and
28 unsupported winner claims. It supports retrospective comparison; confirmation
29 or operational use requires fresh hidden data and separate validation.

Keywords: model comparison, evaluation reliability, benchmark qualification,
dependent-event data, reproducibility, USGS DYFI

1 Introduction

Engineering model selection depends on comparisons that remain valid when a dataset is updated, observations are dependent, and leading scores are close. Without those controls, an apparent improvement can be caused by a changed population, a proxy for the outcome or evaluation axis, cross-split dependence, or unquantified sampling variation rather than by a better model. Standard benchmarking is valuable precisely because it fixes a common task and metrics; recent work in this journal, for example, compares object detectors on COCO to support model selection for engineering applications [30]. The reliability problem is harder for event catalogues because both the catalogue and the dependence structure can evolve.

The U.S. Geological Survey “Did You Feel It?” (DYFI) system is a useful engineering case study. It converts crowdsourced felt reports into a community-decimal intensity (CDI) for each earthquake and is widely reused as an interpreted data product [2, 23, 35, 36]. The target studied here is retrospective classification of whether event-maximum CDI reaches 6 from outcome-blind source metadata. It is not a real-time alert, causal model, or operational triage rule.

Our engineering question is: *do performance and model-superiority claims remain supportable when catalogue version, dependent-event splitting, temporal evaluation, paired uncertainty, and reconstruction are controlled?* The intended users are applied machine-learning engineers evaluating a candidate classifier, maintainers publishing a stable benchmark, and reviewers auditing a claimed improvement. They need two decisions: whether a candidate is admissible for comparison and whether its improvement over a declared reference is supported rather than a point-score fluctuation.

Existing felt-intensity studies use random or stratified partitions and do not provide that reusable decision contract [16, 21].

1.1 Contributions

We address unreliable model comparison by releasing a versioned benchmark and an executable candidate-qualification contract. Concretely, the work provides:

1. **A versioned DYFI benchmark** with a dated, fingerprinted catalogue snapshot, outcome-independent sequence grouping, and a fixed temporal holdout (Sections 3–4.2).
2. **An auditable candidate test** that checks allowed inputs and role isolation before scoring, then uses fixed baselines and paired Brier uncertainty to decide whether an improvement is supported (Sections 4.1–5.5).
3. **A worked model-selection example** showing a clear gain from depth beyond magnitude alone, a competitive two-input logistic baseline, and the point at which a more complex model remains unresolved. Data, frozen parameters, tests, and prediction-reconstruction records are released with the benchmark (Section 5).

The primary contribution is the benchmark and its comparison process, not a new production classifier.

1.2 Engineering significance and intended use

The output is an engineering test harness for transparent retrospective model comparison. A declared candidate enters through the released schema, is excluded if it uses forbidden outcome-derived fields or violates role isolation, is scored on the fixed temporal holdout, and is compared with paired sequence-cluster uncertainty. This workflow prevents an engineer or reviewer from declaring superiority from a different catalogue version, a contaminated split, or an unresolved point difference. Because holdout labels are public, the release is not an untouched prospective test: confirmatory evaluation of

Table 1 Engineering failure modes addressed by the benchmark qualification contract

Failure source	Invalid engineering conclusion	Qualification control
Catalogue revision	Non-identical event population	Frozen access instant + SHA-256 release
Outcome-derived field	Label leakage	F1/F2 schema audit
Identifier or downstream proxy	Proxy leakage	F3/F4 field audit
Sequence crosses roles	Dependent-event contamination	F5 sequence-role audit
Temporal-cutoff violation	Future information in development	F6 temporal audit
Point-score ranking	False model-superiority claim	Paired sequence-cluster interval
Environment drift	Irreproducible scores/artifacts	Pinned environment + reconstruction hash

a new model requires a fresh benchmark version or hidden evaluation service. Table 1 names the failure prevented by each control, and Fig. 1 shows the decision path.

2 Related work and novelty

DYFI and reusable macroseismic data.

Foundational DYFI work defines community decimal intensity, quality control, response dependence, and post-event revision [2, 23, 35, 36]. Rösler et al. [25] therefore recommend preserving timestamped interpreted products. Related resources include INGe and the Western North America macroseismic release [7, 19], historical databases [17, 28], and crowdsourced analyses of shaking and geology [13, 26]. These works motivate a frozen, reusable evaluation resource but do not define the present event-level classification contract.

Machine-learning and engineering comparisons.

Felt-intensity models have used random forests or boosting with instrumental, catalogue, or crowdsourced predictors [6, 15, 16, 20, 21]. The closest same-source study uses a naive split for DYFI regression [11]; a concurrent DYFI benchmark studies

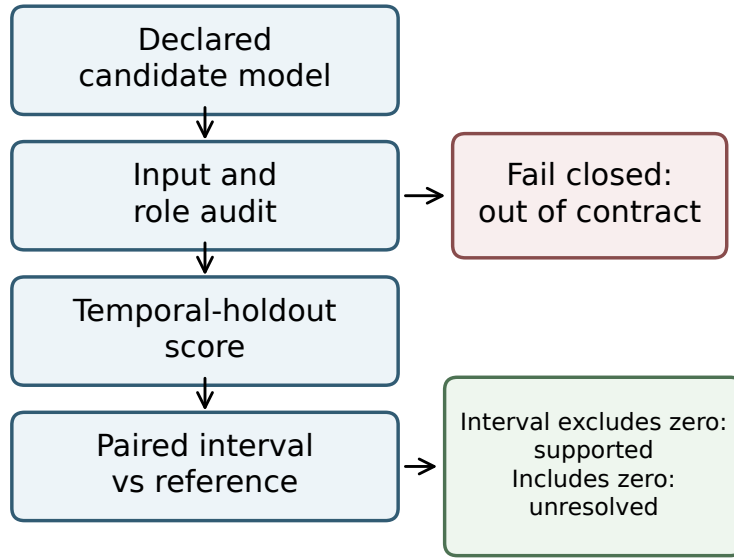


Fig. 1 Candidate-model qualification. A candidate that fails the schema, proxy, role, or temporal audit is inadmissible. An admitted candidate is scored on the fixed holdout and compared with paired uncertainty; an interval including zero remains unresolved rather than becoming a winner claim

per-location MMI simulation and identifier leakage [14]. EarthquakeNPP supplies the nearest broader leakage-aware seismological benchmark but targets occurrence forecasting [29]. Decision-oriented engineering studies also compare tree-based models, including XGBoost, to support bridge-condition forecasting and maintenance planning [5]. Recent engineering studies use benchmark and model comparisons for object detection, rock properties, and pavement maintenance [1, 18, 30], while seismic engineering work makes the affected response and design interpretation explicit [27]. The present study adopts that decision-oriented form for model-comparison reliability rather than proposing a new classifier.

Evaluation-method delta.

Grouped or blocked validation for dependent data and leakage prevention are established principles [12, 24]. The delta is the released combination for this task: a

107 timestamped and hashed single-source corpus, executable feature/proxy and split
108 audits, outcome-independent sequence-grouped roles, a temporal holdout, fixed base-
109 lines, paired sequence-cluster uncertainty, and exact reconstruction. No individual
110 component is claimed as new; the contribution is the auditable engineering comparison
111 contract and its worked decision example.

112 **3 Data, provenance, and outcome-boundary** 113 **disclosure**

114 *Source and single-instant pinning.*

115 The corpus is derived entirely from USGS DYFI and its parent ANSS Comprehen-
116 sive Catalog (ComCat), retrieved through the International Federation of Digital
117 Seismograph Networks (FDSN) event web service [31, 34]. The acquisition envelope
118 is outcome-independent: $M \geq 5$, product type `dyfi`, event type `earthquake`, start-
119 time 2015-01-01 to endtime 2025-01-01. FDSN defines the end bound as inclusive;
120 no frozen record occurs at that boundary, so the realized snapshot spans 2015–2024.
121 Results are ordered by time. Because the catalogue evolves, the data were frozen at
122 one access instant (2026-09-01T06:33:30Z) and assigned a SHA-256 fingerprint. Pre-
123 liminary counts changed from 7,575 to 7,576 during acquisition, documenting a live
124 catalogue revision. The final payload contained 7,520 records, and a count requested
125 50 seconds later also returned 7,520. The earlier counts had no paired payload, so their
126 membership differences cannot be decomposed and are recorded as provenance, not
127 analysis exclusions. The downloaded bytes and their fingerprint define this resource
128 version (Table 2, Fig. 2).

129 ***Protocol chronology.***

130 The file records an internal 2026-09-01 freeze but was first publicly timestamped on
131 2026-09-19. We therefore describe it as an internally frozen protocol, not an externally
132 registered preregistration.

133 ***Retrospective eligibility and post-report ascertainment.***

134 DYFI intensity is an after-the-event quantity whose reliability grows with the number
135 of felt reports [23]. To keep the label meaningful we apply a data-informed eligibility
136 filter of at least ten responses per event. The response count is used *only* for eligibil-
137 ity and is explicitly forbidden as a predictive feature, because it is a post-outcome
138 ascertainment proxy; this forbiddance is enforced by the leakage audit (family F2,
139 Section 4.1). Eligibility yields 2,362 events with severe prevalence 0.298 (703 severe
140 events). Of these, 2,360 events are assigned to the internally specified analysis roles
141 (Section 4.2) and contain 702 severe events; the remaining two are temporal-straddle
142 exclusions, one of which is severe. That single severe straddle event accounts for the
143 difference between the eligible cohort (703 severe) and the analysis-assigned set (702
144 severe), so eligibility is not the same as analysis-role assignment (Table 2).

145 ***Endpoint.***

146 We label an event severe when its event-maximum CDI is at least 6.0 (intensity VI or
147 greater). We use 6.0 as the primary threshold and report sensitivity at 5.5, 6.0, and
148 6.5. We also examine a near-threshold band of ± 0.5 (Section 5.2).

149 ***Outcome-boundary disclosure and held-out semantics.***

150 Only corpus-level outcome-blind counts and inherited aggregate feasibility statistics
151 were exposed before construction; no per-event label informed the endpoint, split, or
152 tuning choices, and role membership is assigned by an internally specified, outcome-
153 independent rule that never uses the label. We therefore describe the external holdout

154 as *held out from model development with aggregate feasibility previously known*, and we
155 deliberately do *not* call it “untouched” or “confirmatory.” Two event identifiers were
156 prelisted for quarantine because their aggregate counts had been seen during source
157 verification; neither identifier occurs in the frozen snapshot, so the rule is non-operative
158 in version 1 and excludes no record. The complete list of protocol decisions is reported
159 in Section 5.8. No respondent-level personally identifying information is redistributed;
160 only event-level source metadata, the severe label, the eligibility-only response count,
161 identifiers, and role assignments are released. Snapshot-level provenance is supplied
162 by the activated source-locator manifest and source payload SHA-256; version 1 does
163 not claim per-event product-version hashes. The release bundle follows datasheet-
164 for-datasets documentation [9] and findable, accessible, interoperable, and reusable
165 (FAIR) data principles [37].

166 4 Benchmark design

167 4.1 Feature sets and the six-category leakage audit

168 The primary *source-only* feature set contains magnitude and depth. It is the only
169 feature set evaluated on the temporal holdout. An *expanded-metadata* sensitivity anal-
170 ysis (baseline B3) additionally uses latitude, longitude, and event year. Event year is
171 known at prediction time and is not intrinsically label leakage, but it is excluded by the
172 declared source-only temporal-holdout contract. The audit accepts only an approved
173 feature set with the expected fields in the expected order; any extra or unknown field
174 is rejected.

175 The executable audit enforces six leakage families. Field-level: F1 direct label or
176 its constituents; F2 post-outcome ascertainment or revision (including the response
177 count, eligibility-only); F3 identifier or source proxy; F4 downstream-product proxy.
178 Split-level: F5 sequence cross-role straddle; F6 temporal-cutoff violation. An axis-
179 variable check additionally rejects origin year under a temporal holdout and, when a

geographic holdout is configured, latitude/longitude/region. The latter audit branch is provided for reuse but was not executed as an external evaluation in this study. This enforces a conservative B3 scope restriction; it is not evidence that event year is a leaking label proxy. The audit is useful independently of model performance: it must reject every deliberately contaminated test case and accept only the approved clean feature set. It is exercised by 110 unit tests across four suites, all passing, and accepted the source-only feature set in the frozen snapshot.

4.2 Sequence grouping and outcome-independent roles

Events are grouped into seismic *sequences* by order-stable single-link space-time connected components (100 km, 30 days); the sequence, not the event, is the unit of role assignment, so events linked by this fixed rule cannot straddle roles. This rule is a deterministic grouping heuristic, not a validated catalogue of all physical mainshock–aftershock relationships; unlinked true aftershocks may therefore remain in different roles. Roles are assigned by an internally specified, outcome-independent rule (fixed assignment key, temporal cutoff year 2023, development fraction 0.2, five cross-validation folds); the rule provably never reads the label. The external holdout contains events on or after 2023. The frozen internal protocol also named a secondary leave-region-out geographic evaluation, but that branch was not executed and no geographic role manifest is claimed. The reported geographic result instead summarizes the temporal-holdout predictions by 10° cell (Supplementary Table S3); Supplementary Table S2 records this deviation. Digital fingerprints of the completed temporal role assignments are published so that third parties can verify them. The partition is exhaustive and disjoint: $1,573 + 357 + 430 + 2 = 2,362$ events, comprising 1,573 train, 357 development, 430 external temporal holdout, and 2 temporal-straddle exclusions (Table 2, Fig. 2). Construction (train plus development) is 1,930 events across 1,530 sequences; the holdout is 430 events across 354 sequences.

206 4.3 Baseline suite and exclusions

207 The fixed suite is B0 no-skill (constant prevalence), B1 magnitude-only logistic regres-
208 sion, B2 source-only logistic regression, B3 expanded-metadata logistic regression
209 (used only in compatible sensitivity analyses), B4 random forest (scikit-learn; 22), and
210 B5 XGBoost [4]. Each learned baseline is tuned *only* by the protocol-defined deter-
211 ministic grouped cross-validation on the construction corpus over a fixed grid, named
212 before evaluation, and carried unchanged to the protected holdout with frozen best
213 parameters. A machine-readable fit log enforces a maximum of 120 model fits; the exe-
214 cuted run used 91. Two comparators are excluded for protocol-defined, reproducible
215 reasons recorded in the protocol-decision table (Section 5.8): B3 is not evaluated
216 on the temporal holdout because origin year is outside the declared source-only fea-
217 ture contract; and the Atkinson–Wald (2007) DYFI intensity-prediction equation [2]
218 is excluded because it cannot be applied to the source-only feature set and tectoni-
219 cally heterogeneous population without inventing a site-distance covariate, a per-event
220 tectonic-regime assignment, and a probability link.

221 4.4 Metrics, calibration, uncertainty, and ranking

222 The primary score is the Brier score [3]; secondary scores are log-loss, AUROC,
223 and calibration intercept/slope [32]. For numerical stability, log-loss clips predicted
224 probabilities symmetrically to $[10^{-15}, 1 - 10^{-15}]$ before taking logarithms. Brier-
225 score uncertainty is quantified by a sequence-cluster bootstrap ($n_{\text{boot}} = 2000$, 95%
226 intervals) that resamples whole sequences rather than events [8], with a minimum-
227 valid-replicate rule. Calibration and feasibility states are *derived*, not hard-coded: a
228 constant-prevalence predictor has no identifiable calibration slope, so the table reports
229 that value as unavailable. Pairwise ranking uses a paired sequence-cluster bootstrap
230 Brier difference with an explicit tie probability; the table reports whether each 95%
231 interval includes zero, together with a tie-aware joint ordering (Section 5.6). Because

the intervals use 2,000 Monte Carlo draws, near-zero endpoints are interpreted cautiously rather than as stable ranking guarantees. The protocol requested intervals for every metric; only Brier-based intervals were retained, so other metrics are descriptive point estimates (Supplementary Table S2).

4.5 Protected evaluation and reproducibility

The official decision-bearing protected evaluation was authorized once by an automated, globally consumed token; the one-shot wrapper refuses a second authorized run. After the result was fixed, the token-free deterministic analytic core was replayed twice, reproducing the recorded result fingerprint exactly. Thus, single-use refers to result authorization, not later verification. The frozen holdout predictions were also recovered bit-for-bit from the released rows and best parameters under a pinned scientific-Python environment (NumPy, SciPy, scikit-learn, XGBoost; 4, 10, 22, 33). All fits, exclusions, subsets, cutoffs, and pairwise comparisons are reported regardless of outcome; the only permitted omissions are the protocol-defined exclusions above. Computation is local-only under a pinned environment.

Reproducible artwork and generative AI assistance.

All figures were generated with Matplotlib 3.11.1 directly from the frozen analysis assets; no manual image manipulation or AI-generated artwork was used. ChatGPT assisted language editing, formatting, code review, literature discovery, and revision suggestions. It did not generate data, run model fits, or create the reported results. The authors verified all sources, computations, interpretations, and assisted changes and retain full responsibility.

5 Results

All result-bearing values below are generated from the frozen analysis outputs and checked automatically against their source records; no result cell was hand-edited.

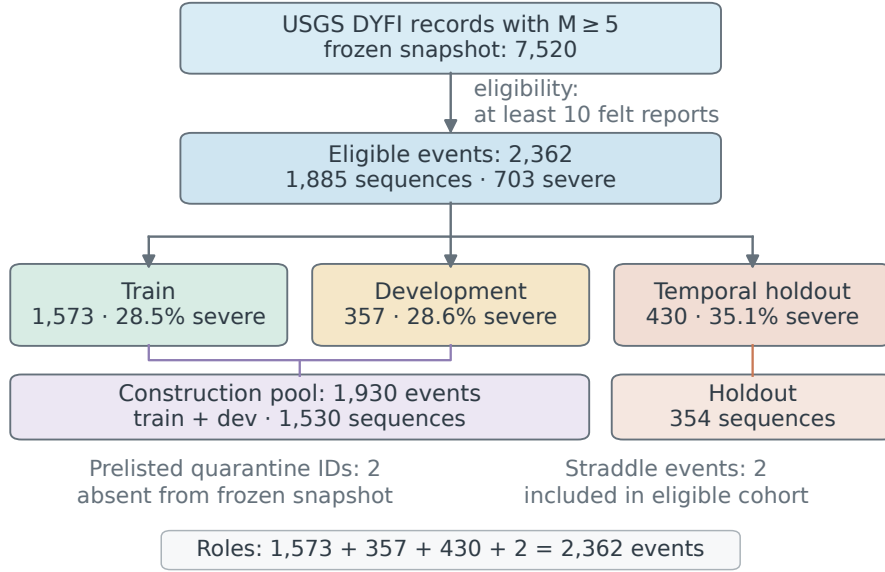


Fig. 2 Cohort construction and role assignment. On the frozen USGS DYFI $M \geq 5$ snapshot. The 7,520 source records yield 2,362 eligible events in 1,885 seismic sequences. Of these, 2,360 are assigned to train ($n = 1,573$), development ($n = 357$), or temporal holdout ($n = 430$); two straddle events are excluded from role assignment

5.1 Cohort and roles

Table 2 and Fig. 2 give the cohort accounting. Of 7,520 frozen DYFI records, 2,362 events pass the ten-response eligibility filter (severe prevalence 0.298; 703 severe events), partitioned into 1,573 train, 357 development, and a 430-event temporal holdout, with two temporal-straddle exclusions. Thus 2,360 of the 2,362 eligible events receive analysis roles; they contain 702 severe events, and the 703rd severe event is one of the two straddle exclusions. Two identifiers prelisted for quarantine are absent from the frozen snapshot, so that safeguard removes no version 1 record. Construction severe prevalence is 0.285 and holdout prevalence is 0.351; the higher holdout prevalence reflects the temporal transport shift and is reported, not adjusted.

Table 2 Cohort accounting and outcome-independent role assignment

Stage	Events	Severe	Severe rate	Sequences
USGS DYFI records with $M \geq 5$ (frozen snapshot)	7520	–	–	–
Eligible (at least 10 felt reports)	2362	703	0.298	1885
Train	1573	449	0.285	–
Development	357	102	0.286	–
Construction (train+dev)	1930	551	0.285	1530
External temporal holdout	430	151	0.351	354
Temporal-straddle excluded	2	1	–	1

Note. Roles were fixed without using outcomes. Two event identifiers prelisted for quarantine are absent from the frozen snapshot, so the safeguard removes no version 1 record. Severe = 1 when an event’s maximum community decimal intensity (CDI) is at least 6. The eligible severe count reconciles as 703 = 449 (train) + 102 (development) + 151 (holdout) + 1 (straddle exclusion). Thus the 2360 role-assigned events contain 702 severe events.

Table 3 Endpoint-threshold probe using fixed source-only probabilities

Maximum CDI cutoff	Severe rate	Brier score	AUROC
5.5	0.458	0.240	0.734
6.0 ^a	0.351	0.181	0.786
6.5	0.260	0.143	0.815

Note. ^a Protocol-defined primary cutoff (≥ 6). CDI denotes community decimal intensity. The sensitivity rows relabel and rescore fixed B2 probabilities; B2 is not refit or recalibrated for the alternate cutoffs. In the construction sample, 0.164 of events lie within 0.5 CDI units of the primary cutoff, and the no-skill Brier score is 0.204. Lower Brier score and higher AUROC indicate better performance.

5.2 Difficulty and endpoint sensitivity

Table 3 and the left panel of Fig. 3 report a descriptive fixed-prediction threshold probe. As the severe cut moves from 5.5 to 6.5, holdout prevalence falls smoothly (0.458, 0.351, 0.260) and the source-only logistic-regression Brier score and AUROC move monotonically (Brier 0.240, 0.181, 0.143; AUROC 0.734, 0.786, 0.815). On the construction corpus the near-threshold fraction ($|\max\text{CDI} - 6| < 0.5$) is 0.164 and the no-skill Brier score is 0.204, bounding the difficulty of the primary endpoint. This probe relabels and rescores probabilities fitted for the primary 6.0 endpoint; it does not refit or recalibrate B2 at 5.5 or 6.5 and therefore does not establish endpoint robustness. The protocol-required aggregation-grid sensitivity was not executed and is disclosed as a deviation (Supplementary Table S2).

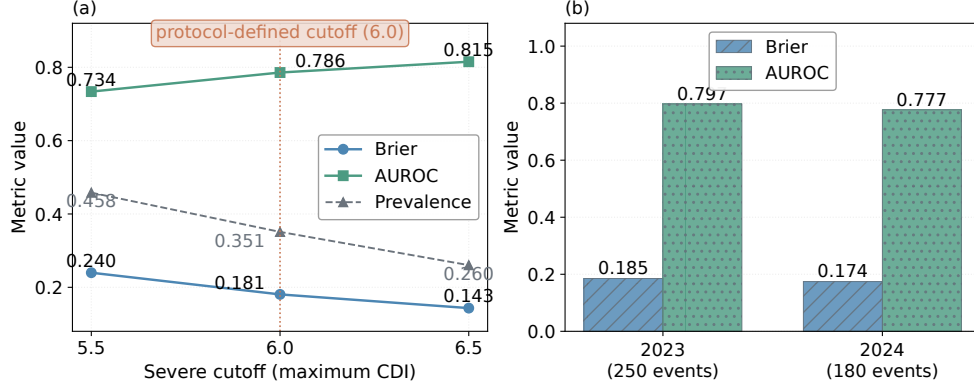


Fig. 3 Endpoint and temporal sensitivity. Results for the source-only logistic regression (B2). (a) Performance as the severe-intensity cutoff varies. (b) Performance within the 2023 ($n = 250$) and 2024 ($n = 180$) portions of the temporal holdout. The cutoff analysis rescores fixed B2 probabilities without refitting. Lower Brier score and higher AUROC are better

5.3 Complementary split-assignment diagnostic

As a secondary stress test, we compare naive random and sequence-grouped folds for a fixed random forest with 300 trees and unrestricted depth under three fold-assignment schemes on the construction corpus (Table 4, Fig. 4). This diagnostic asks whether split assignment changes point performance in this corpus; it is not the qualification workflow’s decision-bearing result. It is also distinct from the tuned temporal-holdout baseline B4, whose maximum depth is 3. On this $M \geq 5$ one-row-per-event universe the point estimates are small and slightly reversed: sequence grouping produced a Brier score 0.0059 lower and an AUROC 0.0124 higher than random-event splitting, calculated from unrounded scores. The sequence-grouped split therefore scores marginally *better* than the naive random split, not worse. The frozen internal protocol required uncertainty reporting for this contrast but did not define a paired interval estimator; the executed result contains point estimates only. We disclose this unfulfilled element and report the contrast descriptively. A plausible reason for the small difference is that the 1,885 sequences are largely singleton (1,530 in construction), leaving little repeated-event structure for grouping to change. Without an interval, however, the result cannot establish a true zero or equivalence. With one row per event, event

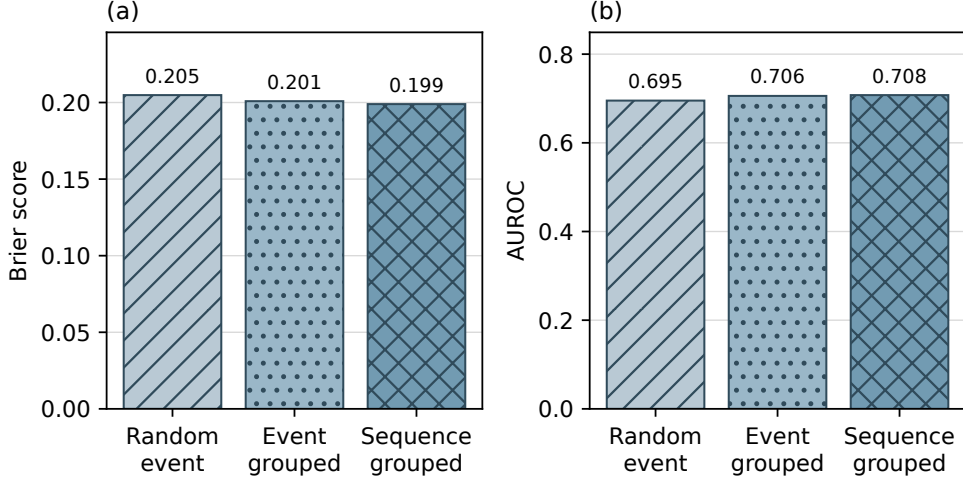


Fig. 4 Split-assignment diagnostic. Five-fold performance of the fixed diagnostic (random forest: 300 trees, unrestricted depth) under random-event, event-grouped, and sequence-grouped fold assignment in the construction sample. Relative to random-event splitting, sequence grouping produced (a) a Brier score 0.0059 lower and (b) an AUROC 0.0124 higher. The diagnostic is distinct from tuned temporal-holdout baseline B4. The small, reversed differences provide no evidence of performance inflation from random splitting in this sample. Both axes start at zero

Table 4 Split-assignment performance of the fixed diagnostic

Fold assignment	Brier score	AUROC	Log-loss	Folds
Random-event (naive)	0.2048	0.6952	0.9640	5
Event-grouped (control)	0.2009	0.7058	0.9223	5
Sequence-grouped (primary)	0.1990	0.7076	0.9709	5

Note. The diagnostic is distinct from the tuned temporal-holdout baseline B4 (depth 3). Using unrounded values, sequence grouping produced a Brier score 0.0059 lower and an AUROC 0.0124 higher than random-event splitting. The frozen internal protocol required uncertainty but did not define the paired interval estimator; the executed report retained point estimates only. This deviation is disclosed, and no post-hoc interval or equivalence claim is added. Because there is one row per event, event grouping imposes the same grouping constraint as random-event assignment, but its deterministic fold allocation is different and can produce different scores.

grouping and random-event assignment impose the same grouping constraint but use different fold allocations. Future applications with more multi-event sequences should repeat this diagnostic with a prespecified paired interval.

5.4 Baseline performance and calibration

Table 5 and Fig. 5 give per-baseline holdout performance ($n = 430$) with sequence-cluster bootstrap 95% intervals. The no-skill B0 has Brier 0.232 ([0.212, 0.253]); its

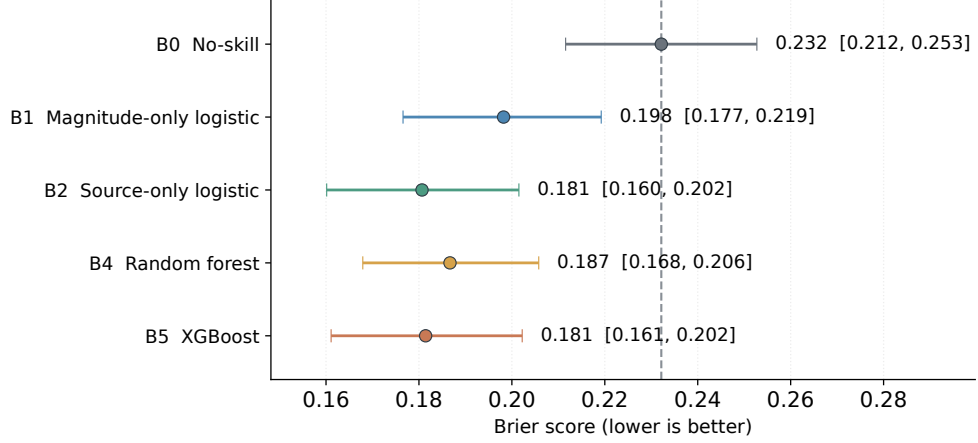


Fig. 5 Temporal-holdout baseline performance. Brier scores and 95% sequence-cluster bootstrap confidence intervals for the five baselines on the temporal holdout ($n = 430$). Lower values are better; the vertical dashed line marks the no-skill B0 score. B3 is omitted because event year is outside the declared source-only temporal-holdout contract (Supplementary Table S2)

Table 5 Temporal-holdout performance of fixed baseline models

Baseline	Brier	95% CI	Log-loss	AUROC	Calibration ^a	Accuracy
B0 (no-skill)	0.232	[0.212, 0.253]	0.658	0.500	INFEASIBLE	0.65
B1 (magnitude-only logistic)	0.198	[0.177, 0.219]	0.583	0.719	0.32 / 1.02	0.70
B2 (source-only logistic)	0.181	[0.160, 0.202]	0.536	0.786	0.64 / 1.29	0.73
B4 (random forest)	0.187	[0.168, 0.206]	0.553	0.777	0.74 / 1.48	0.73
B5 (XGBoost)	0.181	[0.161, 0.202]	0.541	0.783	0.42 / 1.07	0.74

Note. ^a Intercept / slope. These values cannot be estimated for B0 because it assigns the same probability to every event. B3 includes event year and is therefore omitted from the temporal holdout (Supplementary Table S2). Accuracy uses a fixed probability cutoff of 0.5 that was not selected on the holdout. Log-loss clips probabilities to $[10^{-15}, 1 - 10^{-15}]$. Lower Brier score and log-loss are better; higher AUROC and accuracy are better. The two lowest-Brier models cannot be distinguished in their direct paired comparison.

calibration slope cannot be estimated because it assigns the same probability to every event. The learned baselines have lower Brier (B1 0.198, B2 0.181, B4 0.187, B5 0.181) with overlapping intervals; calibration intercept/slope pairs are finite and positive. B2 and B5 both round to the lowest Brier score, 0.181; their paired comparison is reported in Section 5.6.

5.5 Worked candidate qualification

Table 6 applies the released engineering decision rule to the six fixed candidates. B3 is outside the source-only contract because it contains event year; this is a conservative

Table 6 Worked candidate-model qualification using the frozen temporal holdout

Candidate	Gate	Holdout evidence	Decision
B0 no-skill	Pass	Brier 0.232 [0.212, 0.253]; declared reference	Reference
B1 magnitude-only logistic	Pass	Brier 0.198 [0.177, 0.219]; Δ vs B0 -0.0339 [-0.0511, -0.0171]	Supported
B2 source-only logistic	Pass	Brier 0.181 [0.160, 0.202]; Δ vs B0 -0.0515 [-0.0681, -0.0342]	Supported
B4 random forest	Pass	Brier 0.187 [0.168, 0.206]; Δ vs B0 -0.0455 [-0.0598, -0.0307]	Supported
B5 XGBoost	Pass	Brier 0.181 [0.161, 0.202]; Δ vs B0 -0.0507 [-0.0682, -0.0319]	Supported
B3 expanded-metadata logistic	Out of contract	—	Year excluded by schema

Decision rule. A candidate is first admitted by the schema, proxy, role, and temporal audits. A lower paired Brier interval that excludes zero supports improvement over the declared reference; otherwise the comparison remains unresolved. All four learned baselines improve over B0. B2 versus B5 remains unresolved (B2–B5 -0.0008, 95% CI [-0.0078, 0.0068]), so the contract prevents a false winner declaration.

scope exclusion, not demonstrated label leakage. B1, B2, B4, and B5 pass the audit and each has a paired Brier interval below the no-skill B0 reference. The contract therefore supports their improvement over B0. The direct B2–B5 interval includes zero, so the small point difference is not enough to prefer one model on Brier score. This worked example demonstrates the benchmark’s intended use: admit valid candidates, identify supported improvements, and provide a clear reference against which future methods can be tested.

5.6 Pairwise ranking stability

Supplementary Table S1 reports all ten pairwise Brier-score comparisons. Every learned baseline performs better than the no-skill reference B0 (all four intervals exclude zero). The source-only logistic model B2 also improves on the magnitude-only B1 by 0.0176 (95% interval 0.0101–0.0249), showing that depth adds useful predictive information in this dataset. XGBoost B5 improves on the random forest B4, although the lower interval endpoint (0.0003) is near the Monte Carlo boundary. We treat this contrast cautiously and do not use it as a ranking certificate. The B2–B4 and B2–B5 comparisons have intervals that include zero. In the joint bootstrap ordering, B2 has

325 the highest probability of being uniquely best (0.613), followed by B5 (0.387), but the
326 direct B2–B5 comparison remains uncertain. The source-only logistic model therefore
327 provides a strong, simple baseline for future candidates, but the results do not justify
328 naming it a winner.

329 The frozen internal protocol allowed an “equivalent” verdict but did not specify a
330 positive numerical equivalence margin; its resolution floor was zero. We therefore do
331 not invent a post-hoc margin or label any models equivalent. Instead, we report each
332 point difference, 95% interval, and whether that interval includes zero. Supplementary
333 Table S2 records this interpretive restriction, which changes no result.

334 5.7 Temporal and geographic subgroup summaries

335 Supplementary Table S3 and the right panel of Fig. 3 report the temporal and
336 geographic subgroup summaries for the source-only logistic regression within the tem-
337 poral holdout. The two holdout years are internally consistent (2023: $n = 250$, Brier
338 0.185, AUROC 0.797; 2024: $n = 180$, Brier 0.174, AUROC 0.777). The geographic
339 subgroup summary spans 82 occupied 10° cells (per-cell Brier 0.002–0.825, median
340 0.151). These values stratify the same 430 temporal-holdout predictions; they are not a
341 leave-region-out evaluation and support no geographic or cross-system generalization
342 claim.

343 5.8 Exclusions, deviations, and reproducibility

344 Supplementary Table S2 explains the protocol decisions most relevant to interpreta-
345 tion: excluded comparators and records, the unexecuted geographic transport branch,
346 the unfulfilled split-contrast interval, and the absence of an equivalence claim. The
347 tagged public release contains the complete SHA-256 fingerprints, software envi-
348 ronment, and machine-readable verification details. Its manifest fingerprints every

distributed file. The verification record separately identifies three self-referential book-keeping entries and a corrections overlay for three legacy semantic fields; neither changes an analysis file or numerical result.

6 Discussion

Engineering use and decision consequence.

The benchmark gives a researcher a fixed test for a new severe-intensity classifier. It checks inputs and role assignments before scoring and uses a paired Brier interval to judge improvement. The worked example admits four candidates that improve on no-skill, excludes the event-year candidate from the source-only comparison, and identifies the logistic model as the benchmark to beat without forcing a ranking against XGBoost. These outcomes show which comparisons are valid and what a new method must beat. Because all labels are public, confirmation still requires a fresh hidden holdout or independent evaluation service.

Interpreting the split contrast.

The observed point estimate is small and slightly reversed: sequence-grouped folds scored marginally better than random folds. The catalogue contains mostly single-event sequences, so there is little repeated-event structure for grouping to change. Because no interval was retained, the size of this difference remains uncertain. Future applications with more multi-event sequences can use the same diagnostic with a prespecified paired interval.

A practical model-selection result.

All four learned baselines improve on no-skill, and using depth as well as magnitude improves the logistic model. The source-only logistic model and XGBoost both round to Brier 0.181, and their paired comparison is unresolved. The evidence does not show a Brier advantage for the more complex model. The simpler logistic model is therefore

374 a practical default comparator, not a statistically superior winner. Future methods
375 should show a paired improvement over it before added complexity is treated as a
376 performance gain.

377 *Reuse beyond the case study.*

378 The same evaluation pattern can be adapted to other event-based engineering datasets
379 in which related records, changing catalogue versions, or small score differences can
380 make model rankings unreliable. Each application must define its inputs, grouping,
381 holdout, reference, and uncertainty procedure before seeing test results. The compar-
382 ison design is reusable; the present DYFI model is not claimed to transfer to another
383 region, catalogue, or engineering task.

384 **7 Limitations and threats to validity**

385 *Outcome and catalogue limitations.*

386 The endpoint is magnitude-dominated, CDI remains sensitive to responder coverage,
387 and magnitude and depth are frozen preferred catalogue values rather than guaranteed
388 first-alert values. The ten-response rule sets an eligibility and reliability floor but
389 selects on post-event participation and does not remove ascertainment; response count
390 is used only for eligibility. B1 provides the simple magnitude comparator.

391 *Unexecuted sensitivity and transport branches.*

392 The geographic values are temporal-holdout subgroups, not external transport evi-
393 dence. The protocol-defined geographic evaluation and aggregation-grid sensitivity
394 were not executed, and no alternative to the fixed 100-km/30-day sequence defini-
395 tion was tested. The rule guarantees co-assignment only for events it links; it does
396 not guarantee identification of every physical aftershock relationship. Consequently,
397 no geographic, aggregation, sequence-definition, or complete aftershock-isolation
398 robustness claim is made.

399 *Inference and held-out semantics.*

400 The protocol-required split-contrast uncertainty interval was not retained, so that
401 result remains descriptive with no equivalence or zero-effect inference. Non-Brier
402 scores remain descriptive point estimates. Pairwise intervals use 2,000 Monte Carlo
403 draws and near-zero endpoints are interpreted cautiously. The temporal holdout was
404 excluded from model development, but aggregate feasibility was previously known; it
405 is neither “untouched” nor confirmatory.

406 *Public release.*

407 Version 1 of the data, code, frozen results, and reproduction instructions
408 is available in a tagged public repository at [https://github.com/akssha74/](https://github.com/akssha74/dyfi-benchmark-reliability)
409 [dyfi-benchmark-reliability](https://github.com/akssha74/dyfi-benchmark-reliability). The release retains complete digital fingerprints so that
410 later revisions remain distinguishable from the version evaluated here.

411 **8 Conclusion**

412 Reliable engineering model selection requires more than reporting the lowest point
413 score. The released DYFI benchmark makes a candidate pass an auditable schema and
414 role contract, evaluates it on a fixed temporal holdout, and attaches paired uncertainty
415 to the comparison decision. In the worked qualification, all four learned candidates
416 improve over no-skill, and adding depth to the magnitude-only logistic model produces
417 a clear Brier improvement. The source-only logistic model sets a practical benchmark;
418 its comparison with XGBoost remains unresolved. Future models should demonstrate
419 a measurable gain before added complexity is justified. The contract also keeps the
420 event-year candidate outside the declared source-only comparison. The random-versus-
421 sequence diagnostic gives similar point performance in this sparse-sequence corpus and
422 remains descriptive because no interval was retained. All results reconstruct from the
423 frozen release. The contribution is a reusable retrospective test harness for trustworthy

424 model comparison, together with a clear performance bar for future severe-intensity
425 models. Confirmatory use requires a fresh hidden holdout or independent evaluation
426 service; the release is not a real-time warning or operational decision system.

427 **Declarations**

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- 430 • **Competing interests.** The authors have no relevant financial or non-financial
431 interests to disclose.
- 432 • **Ethics approval.** Not applicable. The study analyzes public, aggregate event-level
433 USGS products rather than individual questionnaire responses; it involved no inter-
434 action with participants, identifiable data, human-subject collection, animals, or
435 biological material.
- 436 • **Consent to participate.** Not applicable; no participants were recruited.
- 437 • **Consent for publication.** Not applicable.
- 438 • **Availability of data and materials.** Version 1 data, results, manifests,
439 and instructions are available under CC0 1.0 at [https://github.com/akssha74/](https://github.com/akssha74/dyfi-benchmark-reliability/releases/tag/v1.2.4)
440 [dyfi-benchmark-reliability/releases/tag/v1.2.4](https://github.com/akssha74/dyfi-benchmark-reliability/releases/tag/v1.2.4). Source data are USGS DYFI and
441 ANSS ComCat (public domain; [10.5066/F7J101C8](https://doi.org/10.5066/F7J101C8) and [10.5066/F7MS3QZH](https://doi.org/10.5066/F7MS3QZH)) and
442 are reconstructable via the released fetch-and-verify source-locator manifest. No
443 respondent-level personally identifying information is redistributed.
- 444 • **Availability of code.** Project: DYFI Benchmark Reliability. Homepage: [https://](https://github.com/akssha74/dyfi-benchmark-reliability)
445 github.com/akssha74/dyfi-benchmark-reliability. Archived version: release v1.2.4.
446 Operating systems: platform independent. Programming language: Python. Licence:
447 MIT. Restrictions on non-academic use: none. Full prediction reconstruction and
448 model tests use the separately pinned scientific-Python environment.

- 449 • **Author contributions.** A.S. conceived and designed the benchmark protocol,
450 developed the software, curated the data, performed the analyses, validated the
451 results, and drafted the manuscript. L.P. contributed to study design and inter-
452 pretation, supervised the work, and critically revised the manuscript for important
453 intellectual content. Both authors reviewed and approved the final manuscript and
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457 References

- 458 [1] Alnaqbi A, Zeiada W, Al-Khateeb G (2025) A hybrid machine learning method of
459 support vector regression with particle swarm optimization for predicting IRI in
460 continuously reinforced concrete pavement. Journal of Engineering and Applied
461 Science 72:128. <https://doi.org/10.1186/s44147-025-00706-9>
- 462 [2] Atkinson GM, Wald DJ (2007) “Did you feel it?” intensity data: A surpris-
463 ingly good measure of earthquake ground motion. Seismological Research Letters
464 78(3):362–368. <https://doi.org/10.1785/gssrl.78.3.362>
- 465 [3] Brier GW (1950) Verification of forecasts expressed in terms of probability.
466 Monthly Weather Review 78(1):1–3. [https://doi.org/10.1175/1520-0493\(1950\)
467 078<0001:VOFEIT>2.0.CO;2](https://doi.org/10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2)
- 468 [4] Chen T, Guestrin C (2016) XGBoost: A scalable tree boosting system. In:
469 Proceedings of the 22nd ACM SIGKDD International Conference on Knowl-
470 edge Discovery and Data Mining, pp 785–794, [https://doi.org/10.1145/2939672.
471 2939785](https://doi.org/10.1145/2939672.2939785)
- 472 [5] Elbaroty MG, Zaki MA, Mourad SA (2025) BMS forecasting of bridge health
473 condition degradation using AI machine learning. Civil and Environmental
474 Engineering 21(1):246–253. <https://doi.org/10.2478/cee-2025-0019>

- 475 [6] Ertuncay D, de Lorenzo A, Costa G (2025) Deep learning based earthquake and
476 vehicle detection algorithm. *Journal of Seismology* 29(1):269–281. <https://doi.org/10.1007/s10950-024-10267-8>, first published online in 2024
477
- 478 [7] Fathian Sabet A, Abrahamson N, Parker GA, et al (2026) Macroseismic Intensity
479 Dataset in Western North America. U.S. Geological Survey data release, <https://doi.org/10.5066/P143CZ3R>, URL <https://doi.org/10.5066/P143CZ3R>
480
- 481 [8] Field CA, Welsh AH (2007) Bootstrapping clustered data. *Journal of the Royal*
482 *Statistical Society: Series B (Statistical Methodology)* 69(3):369–390. <https://doi.org/10.1111/j.1467-9868.2007.00593.x>
483
- 484 [9] Gebru T, Morgenstern J, Vecchione B, et al (2021) Datasheets for datasets.
485 *Communications of the ACM* 64(12):86–92. <https://doi.org/10.1145/3458723>
- 486 [10] Harris CR, Millman KJ, van der Walt SJ, et al (2020) Array programming with
487 NumPy. *Nature* 585:357–362. <https://doi.org/10.1038/s41586-020-2649-2>
- 488 [11] Karthik Kumar P, Vismay, Apoorva, et al (2025) Machine learning based DYFI
489 intensity prediction for seismic events in India. In: 2025 9th International Confer-
490 ence on Electronics, Communication and Aerospace Technology (ICECA). IEEE,
491 <https://doi.org/10.1109/iceca66444.2025.11382872>
- 492 [12] Kaufman S, Rosset S, Perlich C, et al (2012) Leakage in data mining: Formulation,
493 detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*
494 6(4):15:1–15:21. <https://doi.org/10.1145/2382577.2382579>
- 495 [13] Kouskouna V, Ganas A, Kleanthi M, et al (2021) Evaluation of macroseismic
496 intensity, strong ground motion pattern and fault model of the 19 july 2019
497 mw5.1 earthquake west of athens. *Journal of Seismology* 25(3):747–769. <https://doi.org/10.1007/s10950-021-09990-3>
498
- 499 [14] Li L, Li D, Ou Z, et al (2025) LLMs as world models: Data-driven and human-
500 centered pre-event simulation for disaster impact assessment. In: *Proceedings of*

- the 2025 Conference on Empirical Methods in Natural Language Processing. Association for Computational Linguistics, Suzhou, China, pp 3078–3096, <https://doi.org/10.18653/v1/2025.emnlp-main.153>, released DYFI-grounded MMI benchmark with a location-identifier leakage test
- [15] Lilienkamp H, Bossu R, Cotton F, et al (2023) Utilization of crowdsourced felt reports to distinguish high-impact from low-impact earthquakes globally within minutes of an event. *The Seismic Record* 3(1):29–36. <https://doi.org/10.1785/0320220039>
- [16] Mori F, Naso G (2025) An explainable machine learning model for predicting macroseismic intensity for emergency management. *Remote Sensing* 17(10):1754. <https://doi.org/10.3390/rs17101754>
- [17] Neefs B, Van Noten K, Vanneste K, et al (2025) The belgian traditional macroseismic (BTM) database of the twentieth century. *Journal of Seismology* 29(5):1005–1036. <https://doi.org/10.1007/s10950-024-10266-9>, first published online in 2024
- [18] Niu L, Cui Q, Luo J, et al (2024) Unconfined compressive strength prediction of rock materials based on machine learning. *Journal of Engineering and Applied Science* 71:137. <https://doi.org/10.1186/s44147-024-00472-0>
- [19] Oliveti I, Faenza L, Michellini A (2022) INGe: Intensity-ground motion data set for Italy. *Annals of Geophysics* 65(1):DM102. <https://doi.org/10.4401/ag-8709>
- [20] Özüpak Y (2026) Explainable and statistically validated machine learning for earthquake alert-level classification using seismic intensity features. *Journal of Seismology* 30(5). <https://doi.org/10.1007/s10950-026-10444-x>
- [21] Patelli L, Cameletti M, De Rubeis V, et al (2026) Machine learning for prompt estimation of macroseismic intensity from seismometric data in Italy. *Scientific Reports* 16:7265. <https://doi.org/10.1038/s41598-026-35740-x>

- 527 [22] Pedregosa F, Varoquaux G, Gramfort A, et al (2011) Scikit-learn: Machine
528 learning in Python. *Journal of Machine Learning Research* 12:2825–2830
- 529 [23] Quitoriano V, Wald DJ (2020) USGS “Did you feel it?”—science and lessons
530 from 20 years of citizen science-based macroseismology. *Frontiers in Earth Science*
531 8:120. <https://doi.org/10.3389/feart.2020.00120>
- 532 [24] Roberts DR, Bahn V, Ciuti S, et al (2017) Cross-validation strategies for
533 data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*
534 40(8):913–929. <https://doi.org/10.1111/ecog.02881>
- 535 [25] Rösler B, Stein S, Hough SE (2022) On the documentation, independence, and
536 stability of widely used seismological data products. *Frontiers in Earth Science*
537 10:988098. <https://doi.org/10.3389/feart.2022.988098>
- 538 [26] Saraò A, Tamaro A, Sandron D, et al (2023) On the crowdsourcing of macro-
539 seismic data to characterize geological settings. *International Journal of Disaster*
540 *Risk Reduction* 96:103934. <https://doi.org/10.1016/j.ijdr.2023.103934>
- 541 [27] Seleemah MA, Helam MS, Abu-alenein MA, et al (2022) Response of Aswan cable-
542 stayed bridge to spatial non-synchronous seismic excitations. *Journal of Engineer-*
543 *ing and Applied Science* 69:70. <https://doi.org/10.1186/s44147-022-00124-1>
- 544 [28] Sira C, Schirr L, Bertrand V, et al (2025) MACROSISDATA: Safeguarding
545 and promoting the heritage of french macroseismic surveys carried out between
546 1921 and 1996. *Journal of Seismology* 29(5):1145–1160. <https://doi.org/10.1007/s10950-025-10322-y>
547
- 548 [29] Stockman S, Lawson DJ, Werner MJ (2026) EarthquakeNPP: A benchmark for
549 earthquake forecasting with neural point processes. *Transactions on Machine*
550 *Learning Research* URL <https://openreview.net/forum?id=dIcNAg6ZuZ>, article
551 6105

- 552 [30] Tian J, Jin Q, Wang Y, et al (2024) Performance analysis of deep learning-
553 based object detection algorithms on COCO benchmark: A comparative study.
554 Journal of Engineering and Applied Science 71:76. <https://doi.org/10.1186/s44147-024-00411-z>
555
- 556 [31] U.S. Geological Survey, Earthquake Hazards Program (2017) Advanced national
557 seismic system (ANSS) comprehensive catalog of earthquake events and products.
558 U.S. Geological Survey, <https://doi.org/10.5066/F7MS3QZH>
- 559 [32] Van Calster B, McLernon DJ, van Smeden M, et al (2019) Calibration: The
560 Achilles heel of predictive analytics. BMC Medicine 17:230. <https://doi.org/10.1186/s12916-019-1466-7>
561
- 562 [33] Virtanen P, Gommers R, Oliphant TE, et al (2020) SciPy 1.0: Fundamental
563 algorithms for scientific computing in Python. Nature Methods 17:261–272. <https://doi.org/10.1038/s41592-019-0686-2>
564
- 565 [34] Wald DJ, Dewey JW (2005) Did You Feel It? U.S. Geological Survey data release,
566 <https://doi.org/10.5066/F7J101C8>
- 567 [35] Wald DJ, Quitoriano V, Dengler LA, et al (1999) Utilization of the Internet for
568 rapid community intensity maps. Seismological Research Letters 70(6):680–697.
569 <https://doi.org/10.1785/gssrl.70.6.680>
- 570 [36] Wald DJ, Quitoriano V, Worden CB, et al (2012) USGS “Did you feel it?”
571 internet-based macroseismic intensity maps. Annals of Geophysics 54(6):688–707.
572 <https://doi.org/10.4401/ag-5354>
- 573 [37] Wilkinson MD, Dumontier M, Aalbersberg IJ, et al (2016) The FAIR guid-
574 ing principles for scientific data management and stewardship. Scientific Data
575 3:160018. <https://doi.org/10.1038/sdata.2016.18>